

GNU-Plot-Lab-Notebook

Suparnajit Ghosh

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Contents

Section 1	4
Question 1	4
Question 2	6
Question 3	8
Question 4	10
Question 5	12
Question 6	14
Question 7	16
Section 2	18
Question 1	18
Question 2	20
Question 3	22
Question 4	24
Question 5	26
Question 6	28
Question 7	30
Section 3	32
Question 1	32
Question 2	34
Question 3	36
Question 4	38
Question 5	40
Question 6	42

Section 1

Question 1

Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set 1000 samples. Take the range of y -values from -4 to $+4$. Mark each curve with different line styles and line widths. Label x -axis and y -axis as 'x-values' and 'y-values' respectively. Set the title of the plot as 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xxi_ou.svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [-10:10]
10 set yrange [-4:4]
11 set xlabel "x-values"
12 set ylabel "y-values"
13 set samples 1000
14
15 f1(x)=(1/(tanh(x)))-(1/x)
16 f2(x)=exp(-sin(x))
17 f3(x)=x*abs(x)
18
19 plot f1(x) lw 1 lc "#5f0f40" dt 1, f2(x) lw 2 lc "#fb8b24" dt 2, f3(x) lw 3 lc "#9a031e" dt 3
```

The output plot is given in the following page.

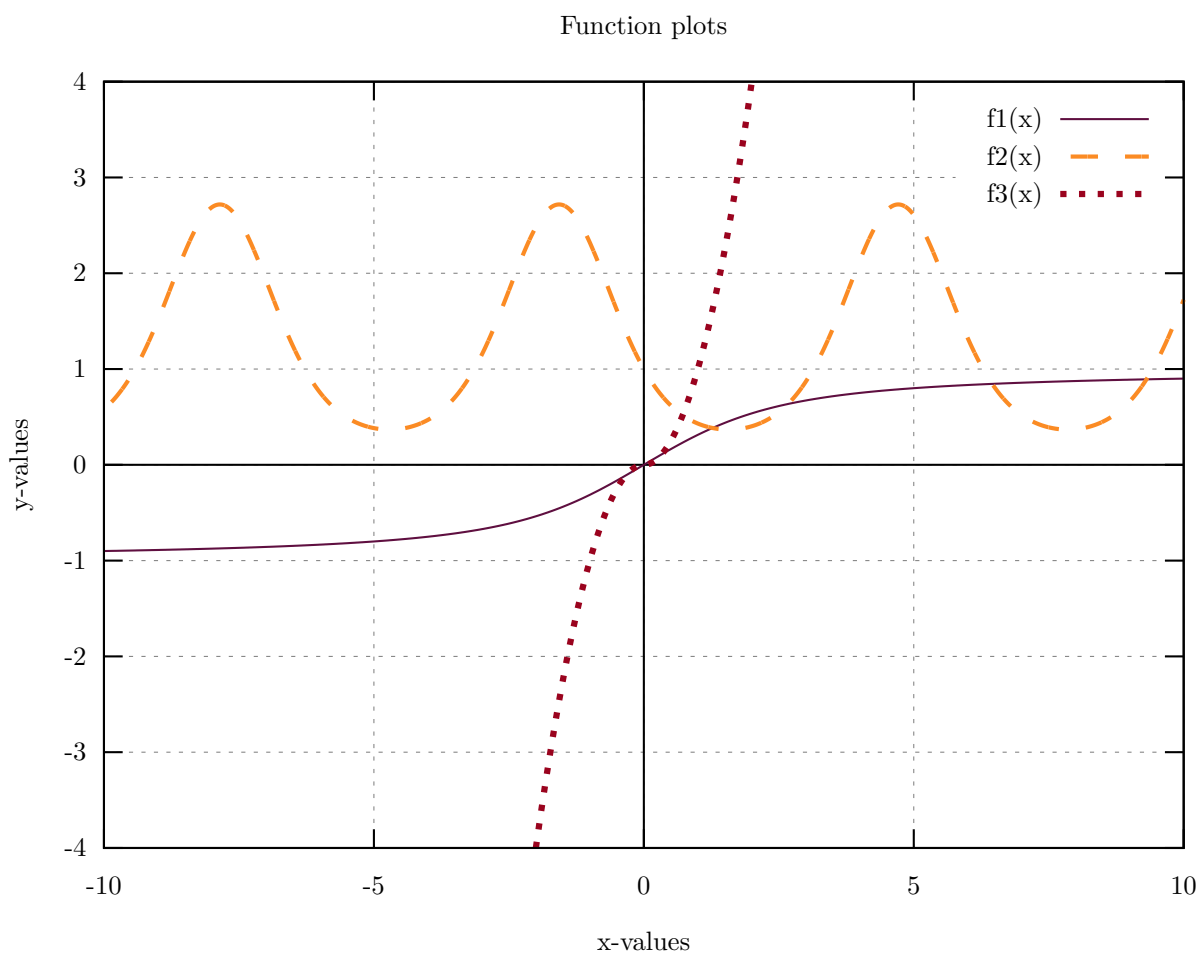


Figure 1: $f_1(x) = \coth(x) - \frac{1}{x}$, $f_2(x) = e^{-\sin(x)}$, $f_3(x) = x|x|$

Question 2

Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2}, \quad f_3(x) = 12.34x^{5/3}$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xx_ou.svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [0:pi]
10 set yrange [0:8]
11 set xtics 0.5
12 set ytics 1
13 set xlabel "x-values"
14 set ylabel "y-values"
15
16 f1(x)=sin(sqrt(x))*2
17 f2(x)=exp(-x/2)
18 f3(x)=12.34*(x**(5/3))
19
20 set samples 1000
21 plot f1(x) lw 1 dt 1 lc "#780000", f2(x) lw 2 dt 2 lc "#c1121f", f3(x) lw 3 dt 3
    lc "#003049"
```

The output plot is given in the following page.

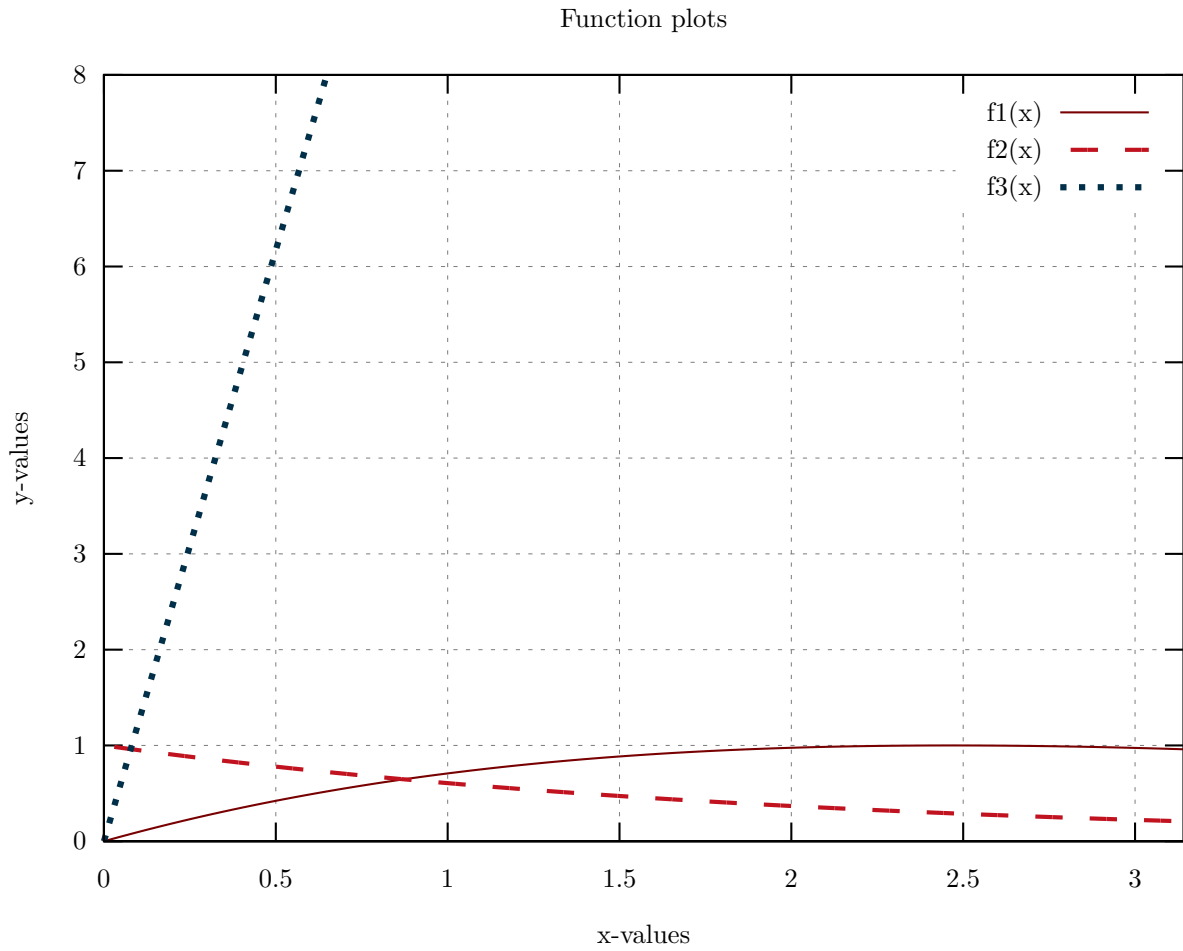


Figure 2: $f_1(x) = \sin^2(\sqrt{x})$, $f_2(x) = e^{-x/2}$, $f_3(x) = 12.34x^{5/3}$

Question 3

Define three functions in the given range:

$$\begin{aligned} f_1(x) &= \ln\left(\frac{1}{x}\right), & x > 0.1, \\ f_2(x) &= 3x^2 \sin x, & -1 < x < 0.1, \\ f_3(x) &= \cos 3x, & x < -1. \end{aligned}$$

Set the range of x values from -3 to $+3$ and the range of y values from -2.5 to $+2.5$. Show both the axes on the plot. Set y label "Piecewise continuous function". Set ticks of the y axis at an interval of 0.5 . Plot the three functions and store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiv_ou.svg"
4
5 set grid
6
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9
10 set ytics 0.5
11
12 set xrange [-3:3]
13 set yrange [-2.5:2.5]
14
15 set ylabel "Piecewise continuous function"
16
17
18 f1(x)=(x>0.1)?(log(1/x)):NaN #Not as Number
19 f2(x)=(x<0.1 && x>-1)?(3*(x**2)*sin(x)):NaN
20 f3(x)=(x<-1)?(cos(3*x)):NaN
21
22 plot f1(x) lc "#9b2226" lw 2 , f2(x) lc "#bb3e03" lw 2 , f3(x) lc "#ee9b00" lw 2
```

The output plot is given in the figure below.

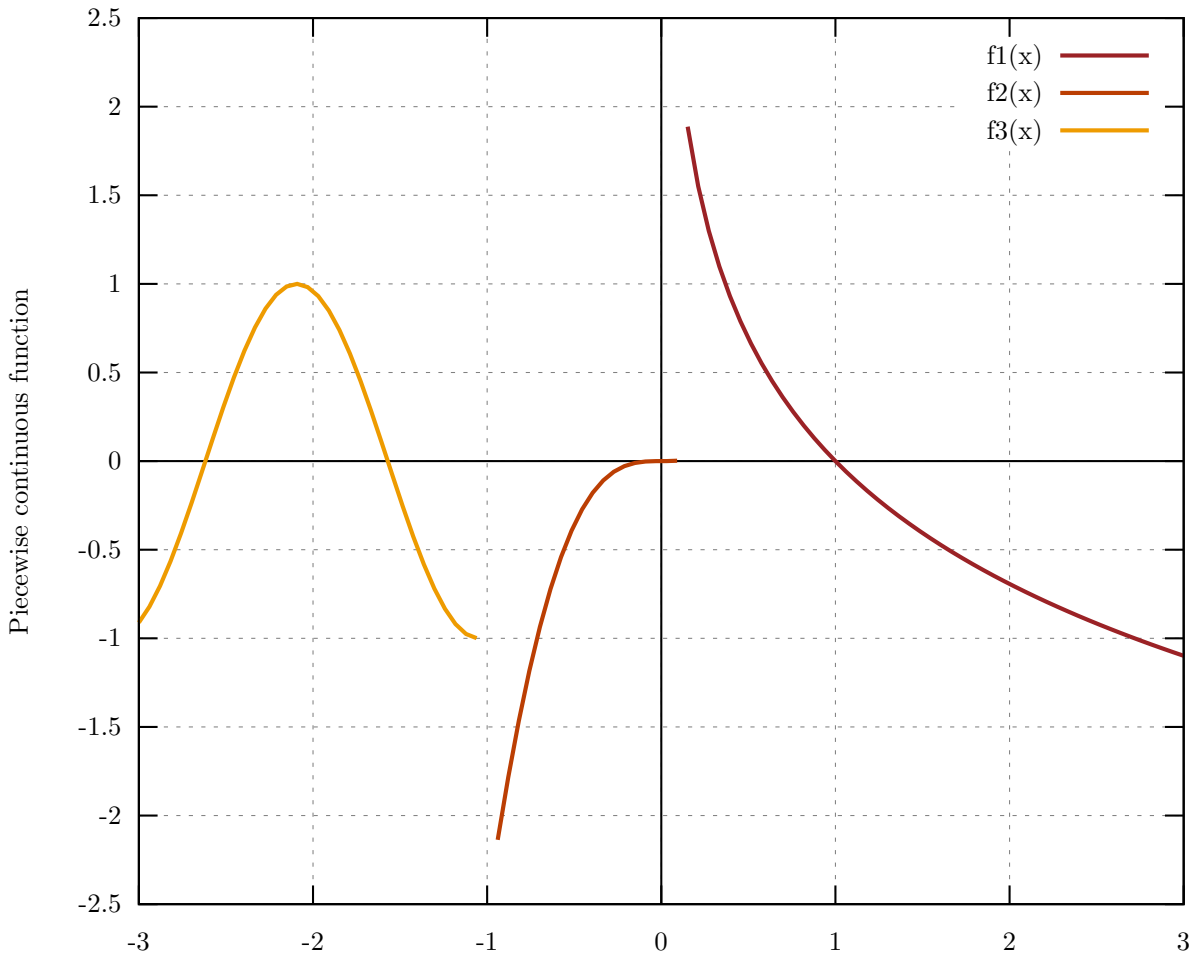


Figure 3: Plot of the piecewise continuous functions $f_1(x)$, $f_2(x)$, and $f_3(x)$

Question 4

Plot

$$2x \sin 2x, \quad \frac{1}{2}x \sinh\left(\frac{x}{2}\right)$$

on the same graph. The line width of the two curves to be different. Set the range of x values from $-\pi$ to $+\pi$ and the range of y values from -5 to 5 . Show both the axes on the plot. Ticks of x axis will be at interval of 0.5 . The label of x and y axis to be "x values" and "y values" respectively. Store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xvi_ou.svg"
4
5 set grid
6 set xzeroaxis linestyle 1 lc "black"
7 set yzeroaxis linestyle 1 lc "black"
8 set xrange [-pi:pi]
9 set yrange [-5:5]
10 set xtics 0.5
11 set xlabel "x values"
12 set ylabel "y values"
13
14 f1(x)=2*x*sin(2*x)
15 f2(x)=(x/2)*sinh(x/2)
16
17 plot f1(x) lc "#9a031e" lw 2, f2(x) lc "#0f4c5c" lw 3
```

The output plot is given in the figure below.

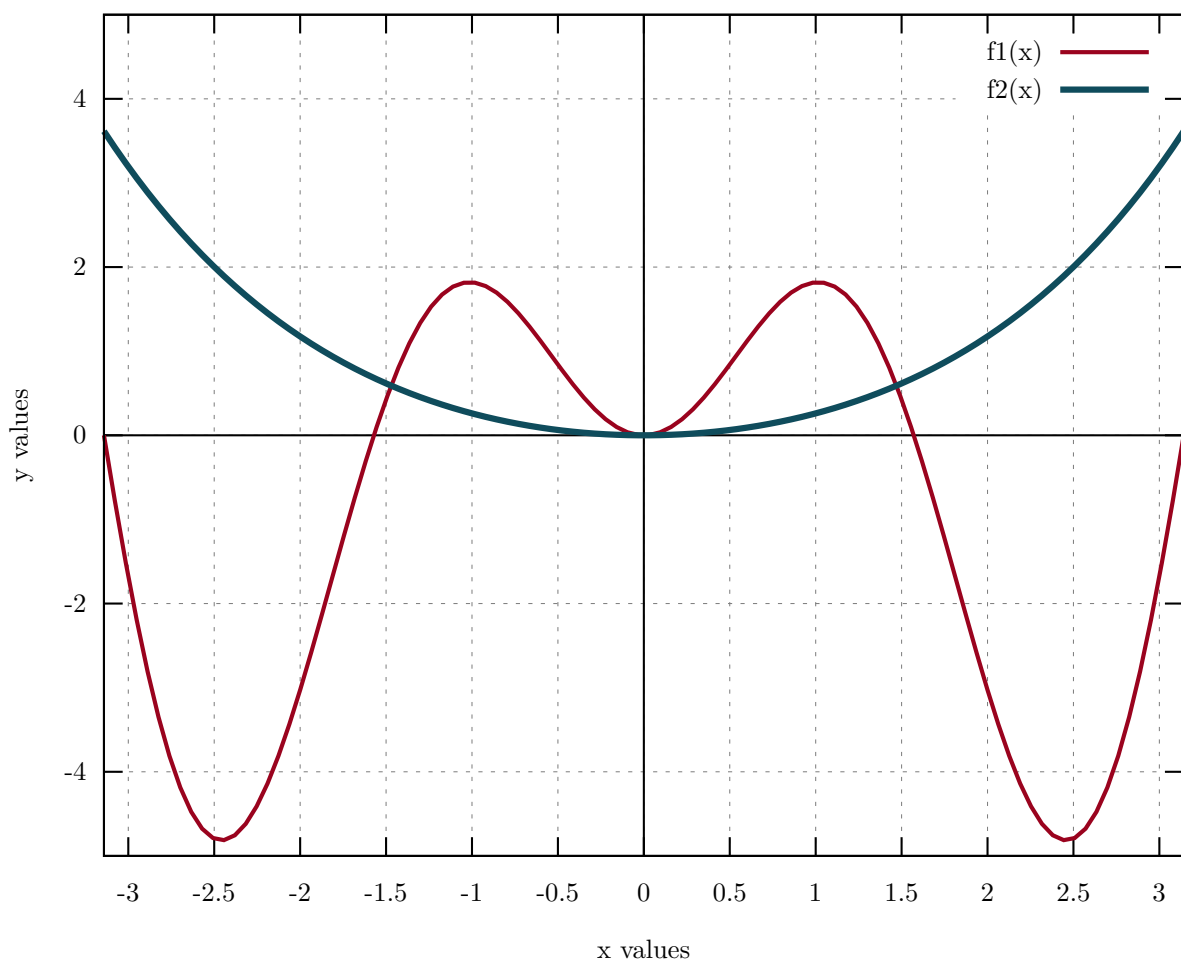


Figure 4: $2x \sin 2x$, $\frac{1}{2}x \sinh\left(\frac{x}{2}\right)$

Question 5

Problem Set 1 (iii). Plot $\sin(x)$ and $2\sin(-x)$ simultaneously on the same plot within the range -3π to 3π . Use the system-defined value of 3π as `3*pi`. Label the x -axis as “x values” and the y -axis as “standing wave”. Save the output plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_iii_ou.svg"
4
5 set grid
6 set xzeroaxis linetype 1 lc "black"
7 set yzeroaxis linetype 1 lc "black"
8 set xrange [-3*pi:3*pi]
9 set yrange [-3:3]
10 set xlabel "x values"
11 set ylabel "standing wave"
12
13 plot sin(x) lc "orchid4" lw 2, 2*sin(-x) lw 2
```

The output plot is given in the figure below.

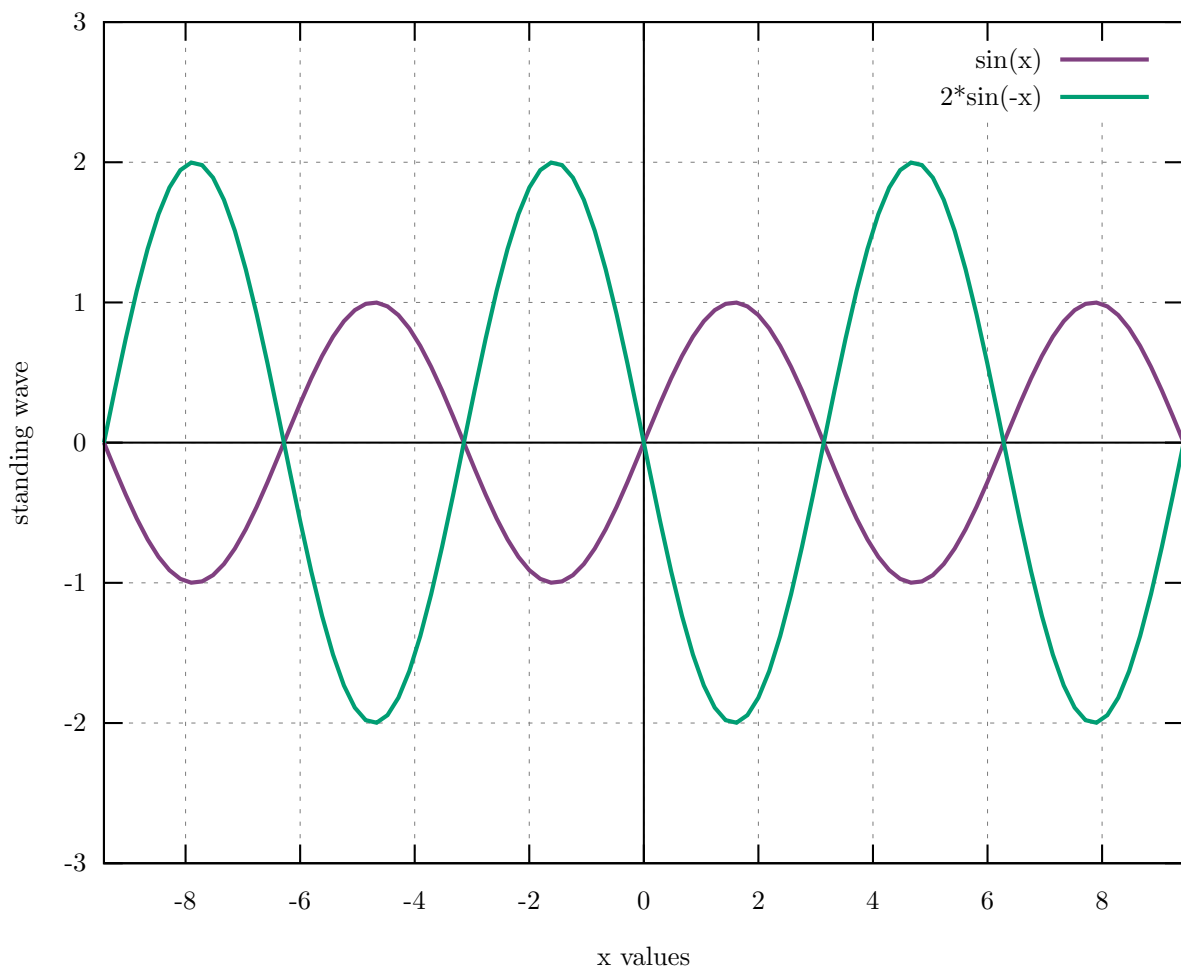


Figure 5: The functions $\sin(x)$ and $2\sin(-x)$ plotted over $-3\pi \leq x \leq 3\pi$.

Question 6

Problem Set 1 (xiii). Define the following piecewise function:

$$f(x) = \begin{cases} x \cos x, & x < 0, \\ x^2 + 3x, & 0 \leq x \leq 1, \\ \ln x + 4, & x > 1. \end{cases}$$

Plot it for $-3 \leq x \leq 3$ and $-2 \leq y \leq 5$. Show both axes and set the title to “Continuous function without derivative at all points”. Save the plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiii_ou.svg"
4
5 set title "Continuous function without derivative at all points"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [-3:3]
10 set yrange [-2:5]
11
12 f(x)=(x<0)?(x*cos(x)):(x>=0 && x<=1)?(x**2 + 3*x):(x>1)?(log(x)+4):NaN
13
14 plot f(x) lc "#ee6c4d" lw 2
```

The output plot is given in the figure below.

Continuous function without derivative at all points

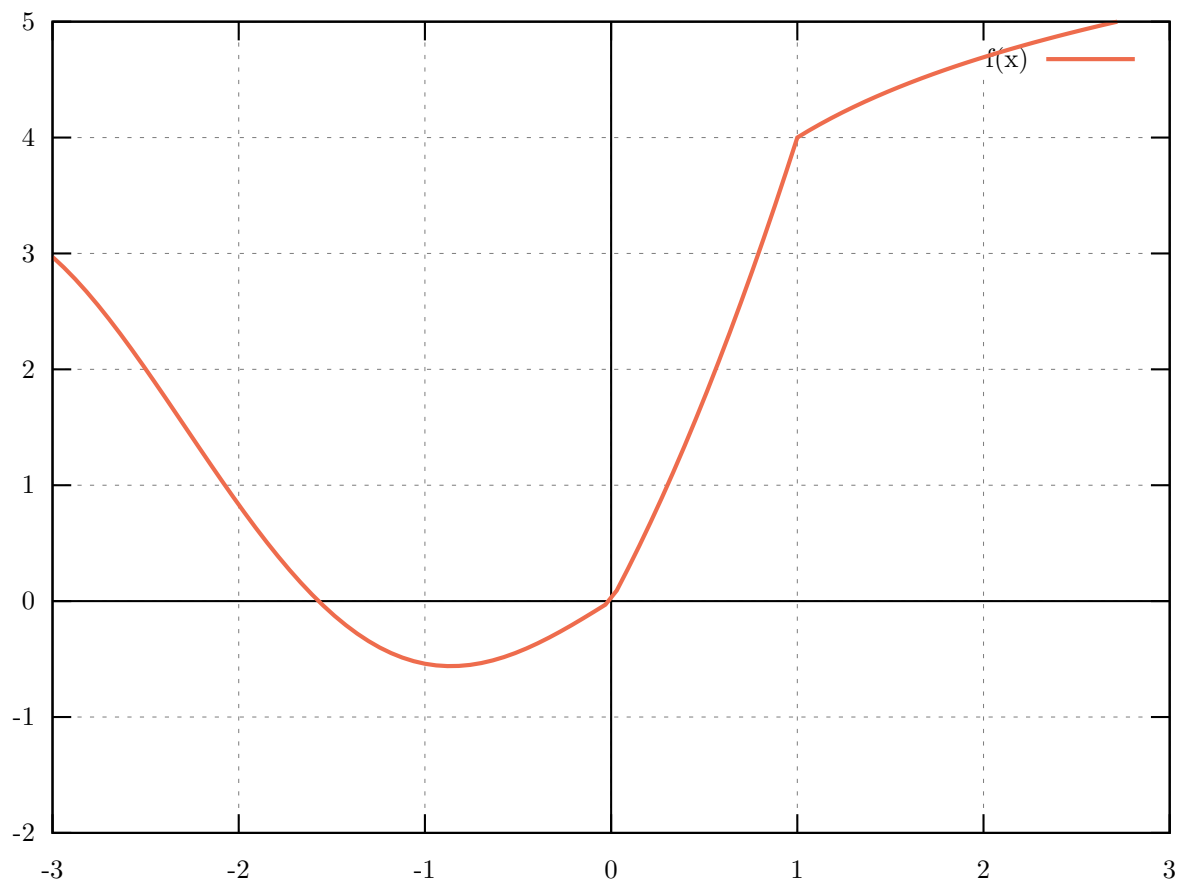


Figure 6: A continuous piecewise function with changes in derivative.

Question 7

Problem Set 1 (xii). Plot $\frac{\sin x}{x}$, $\frac{\sinh x}{x}$, and $y = 1$ on the same graph. Set the x -range from $-\pi$ to $+\pi$ and the y -range from 0 to 5. Show the y -axis, set the title to “Plot of different functions of x ”, and label the axes “ x values” and “ y values”. Use different line widths for all three curves and save the plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xii_ou.svg"
4
5 set grid
6 set title "Plot of different functions of x"
7 set xrange [-pi:pi]
8 set yrange [0:5]
9 set xtics 0.5
10 set ytics 1
11 set xlabel "x values"
12 set ylabel "y values"
13 set xzeroaxis linestyle 1 lc "black"
14 set yzeroaxis linestyle 1 lc "black"
15
16 y1(x)=(x==0)?1:sin(x)/x
17 y2(x)=(x==0)?1:sinh(x)/x
18 y3(x)=1
19
20 plot y1(x) lc "#6F1D1B" lw 1, y2(x) lc "#fb8500" lw 2, y3(x) lc "#f94144" lw 3
```

The output plot is given in the figure below.

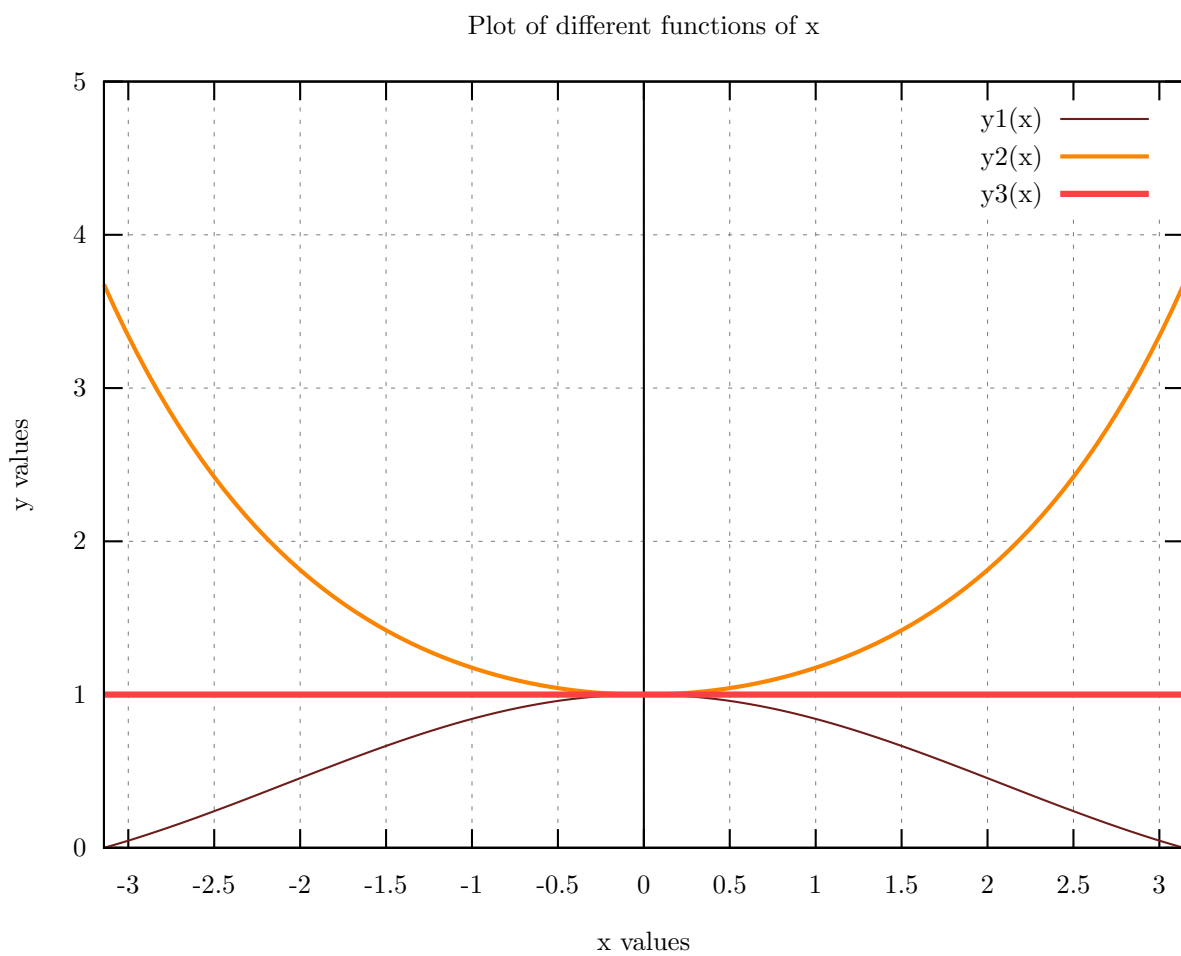


Figure 7: The functions $\frac{\sin x}{x}$, $\frac{\sinh x}{x}$, and $y = 1$.

Section 2

Question 1

Problem Set 2 (i). Fit the data with $y = mx + c$ over $-4 \leq x \leq 8$. Plot the data and fitted line, choosing the y -range so the intercept c is visible on the y -axis. Report m and c .

Table 1: Data for Problem Set 2 (i).

x	y
-2.7	-15.07
-1.5	-9.5
1.4	3.72
3.3	12.5
6.1	25.2

Data file contents:

```
-2.7 -15.07
-1.5 -9.5
1.4 3.72
3.3 12.5
6.1 25.2
```

Sol: The fitted line and data points are shown below; the fit results follow the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_i_ou.svg"
4
5 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_i.log"
6
7 set grid
8 set xzeroaxis linestyle 1 lc "black"
9 set yzeroaxis linestyle 1 lc "black"
10
11 set xrange [-4:8]
12
13 # Fitting-curve
14
15 y(x)=m*x + c
16
17 # Initial guess
18 m=1
19 c=-1
20
21 #Fitting the curve
22 fit y(x) "Lab-files/Data/2/Q_i.txt" using 1:2 via m,c
23
24 #Plotting
25 plot "Lab-files/Data/2/Q_i.txt" using 1:2 with p ps 1.5 title "Data points", \
26      y(x) lw 2 lc "orange" title "Fitted line"
```

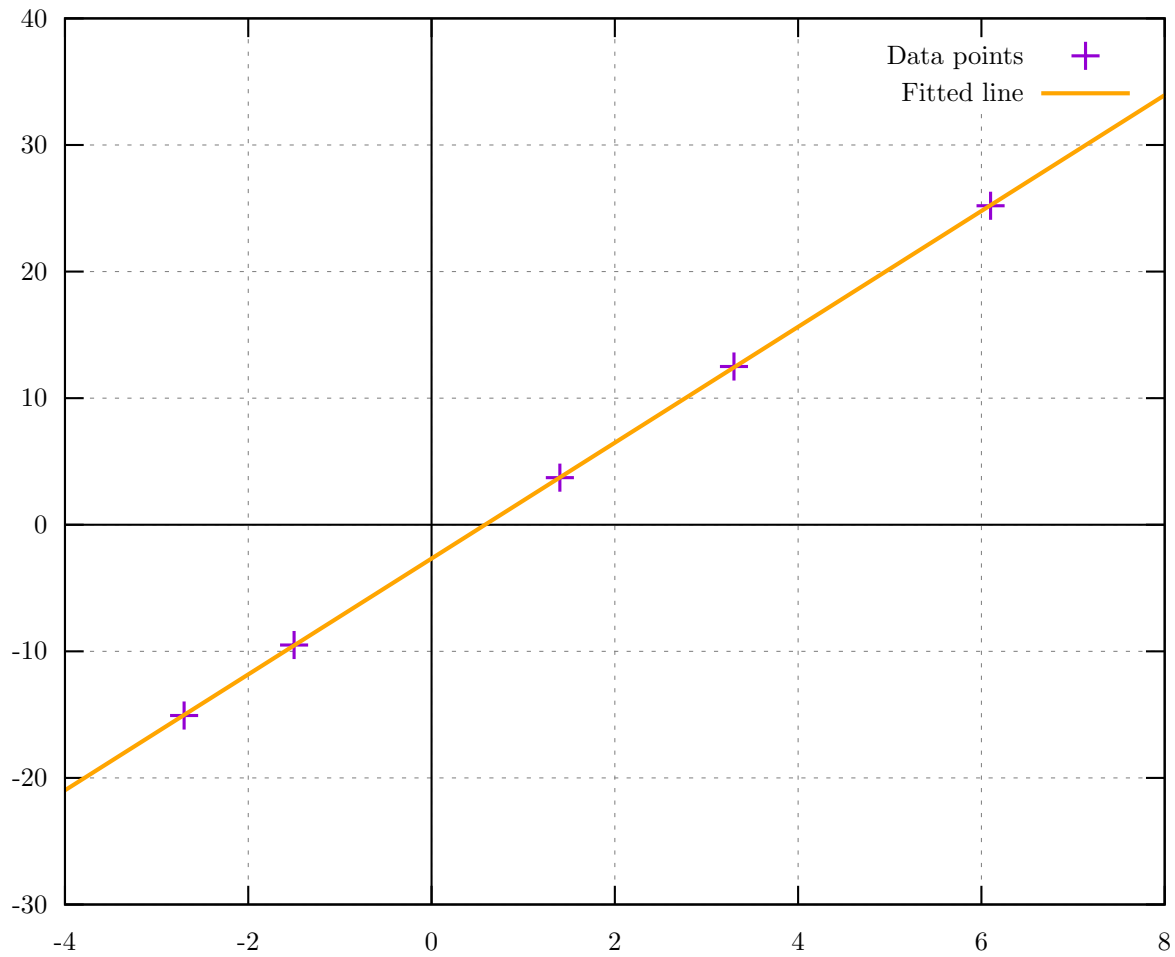


Figure 8: Linear fit for Problem Set 2 (i).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 4.57623	+/- 0.008141	(0.1779%)
c	= -2.67062	+/- 0.02811	(1.052%)

Question 2

Problem Set 2 (ii). Fit the data with $y = mx + c$ over $-7 \leq x \leq 7$. Plot the points and fitted line, show the y -axis, and choose the y -range so the intercept c is visible on it. Report m and c .

Table 2: Data for Problem Set 2 (ii).

x	y
-5.1	21.8
-3.4	16.12
-0.5	7.05
1.57	0.42
6.2	-14.2

Data file contents:

```
-5.1 21.8
-3.4 16.12
-0.5 7.05
1.57 0.42
6.2 -14.2
```

Sol: The data and fitted line are plotted below; the fit results follow the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_ii_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_ii.log"
7
8
9 set grid
10 set xzeroaxis linestyle 0 lc "black"
11 set yzeroaxis linestyle 0 lc "black"
12
13 set xrange [-7:7]
14 #Fitting_curve
15 y(x)=m * x + c
16
17 #Initial guess
18 m=-1
19 c=5
20 #Curve fitting
21 fit y(x) "Lab-files/Data/2/Q_ii.txt" using 1:2 via m,c
22 #Plotting
23 plot "Lab-files/Data/2/Q_ii.txt" using 1:2 with p ps 1.5 lc "#fb8b24" title "Data
    points", \
24     y(x) lw 2 lc "#9a031e" title "Fitted line"
```

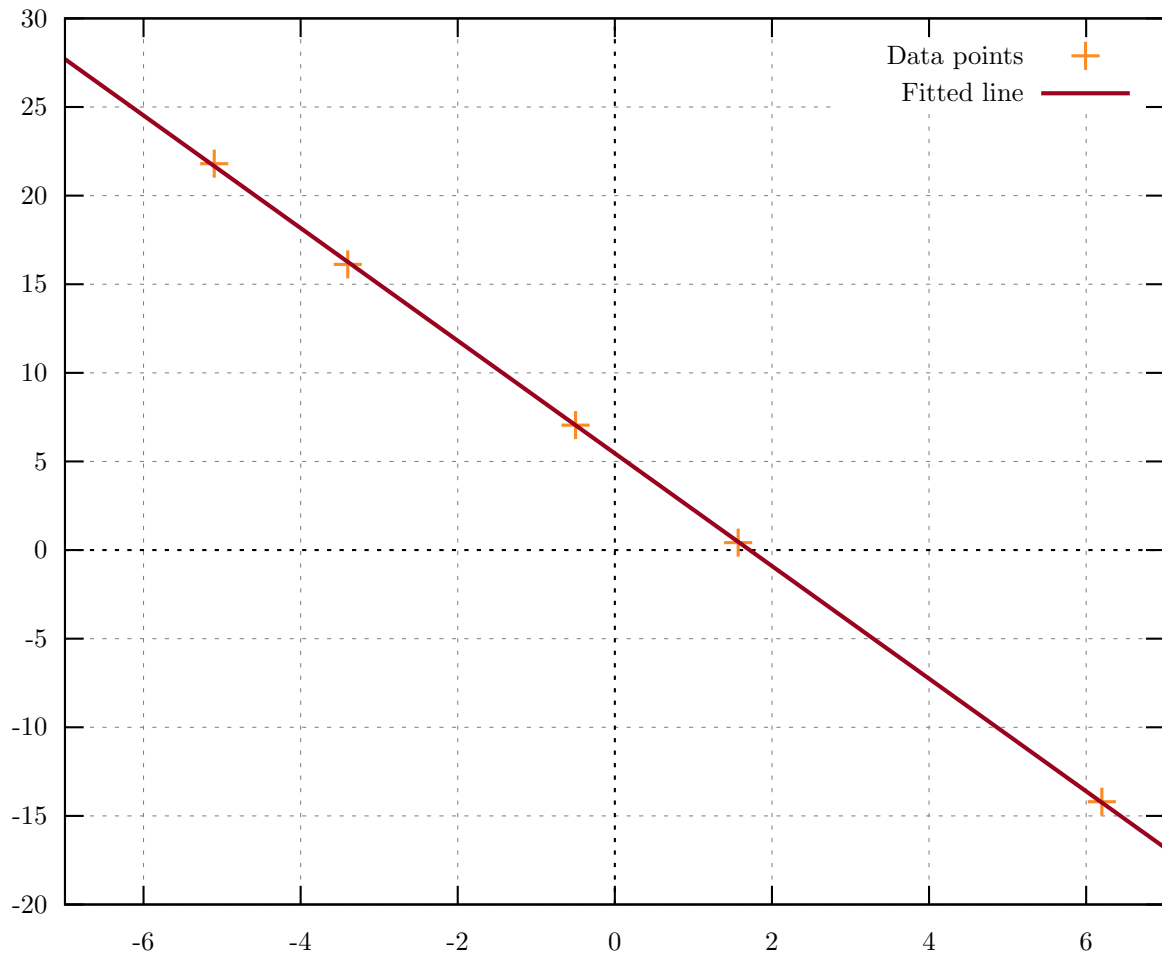


Figure 9: Linear fit for Problem Set 2 (ii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= -3.17793	+/- 0.01351	(0.4251%)
c	= 5.45623	+/- 0.05361	(0.9825%)

Question 3

Problem Set 2 (iii). Fit the data with $y = mx + c$ over $0 \leq x \leq 9$. Plot the points and fitted line, show the y -axis, and choose the y -range so the intercept c is visible on it. Report m and c .

Table 3: Data for Problem Set 2 (iii).

x	y
2.1	-2.5
4.6	7.7
5.7	12.17
6.6	15.9
8.2	22.4

Data file contents:

```
2.1 -2.5
4.6 7.7
5.7 12.17
6.6 15.9
8.2 22.4
```

Sol: The data and fitted line are plotted below; the fit results follow the plot.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_iii_ou.svg"
4
5
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_iii.log"
7
8  set grid
9  set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11 set xrange [0:9]
12
13 #Fitting Curve
14
15 y(x)= m * x + c
16
17 #Initial Guess
18 c=-10
19 m=2
20 #Curve Fitting
21 fit y(x) "Lab-files/Data/2/Q_iii.txt" using 1:2 via m,c
22 #Plotting
23
24 plot "Lab-files/Data/2/Q_iii.txt" using 1:2 with p lc "#e76f51" ps 1.5 title "Data
    points", \
25     y(x) lw 2 lc "#621708" title "Fitted line"
```

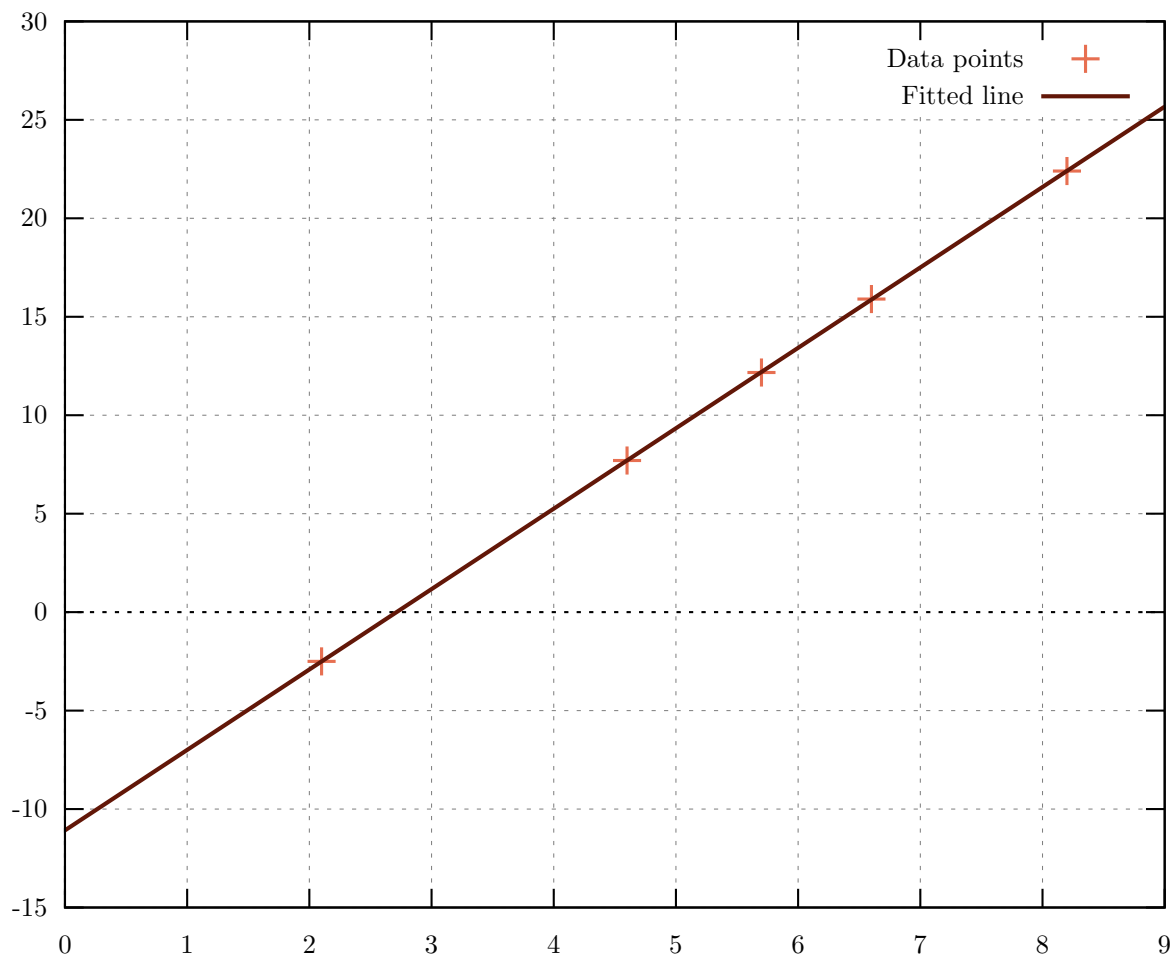


Figure 10: Linear fit for Problem Set 2 (iii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 4.08358	+/- 0.005003	(0.1225%)
c	= -11.0807	+/- 0.02907	(0.2624%)

Question 4

Problem Set 2 (iv). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 4$. Plot the points and fitted line, show both axes, and choose the y -range so the intercept c is visible on the y -axis. Report m and c .

Table 4: Data for Problem Set 2 (iv).

x	y
-2.0	-4.15
-1.3	-2.40
1.8	6.60
2.5	8.98
3.6	11.94

Data file contents:

```
-2.0 -4.15
-1.3 -2.40
1.8 6.60
2.5 8.98
3.6 11.94
```

Sol: The data and fitted line are plotted below; the fit results follow the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_iv_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_iv.log"
7
8 set grid
9 set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set xrange [-3:4]
13
14 y(x)=m*x+c
15 c=0.1
16 m=1
17
18 fit y(x) "Lab-files/Data/2/Q_iv.txt" using 1:2 via m,c
19
20 plot "Lab-files/Data/2/Q_iv.txt" using 1:2 ps 1.5 lc "#ff6b35" title "Data points"
21 , \
    y(x) lw 2 lc "#1a659e" title "Fitted line"
```

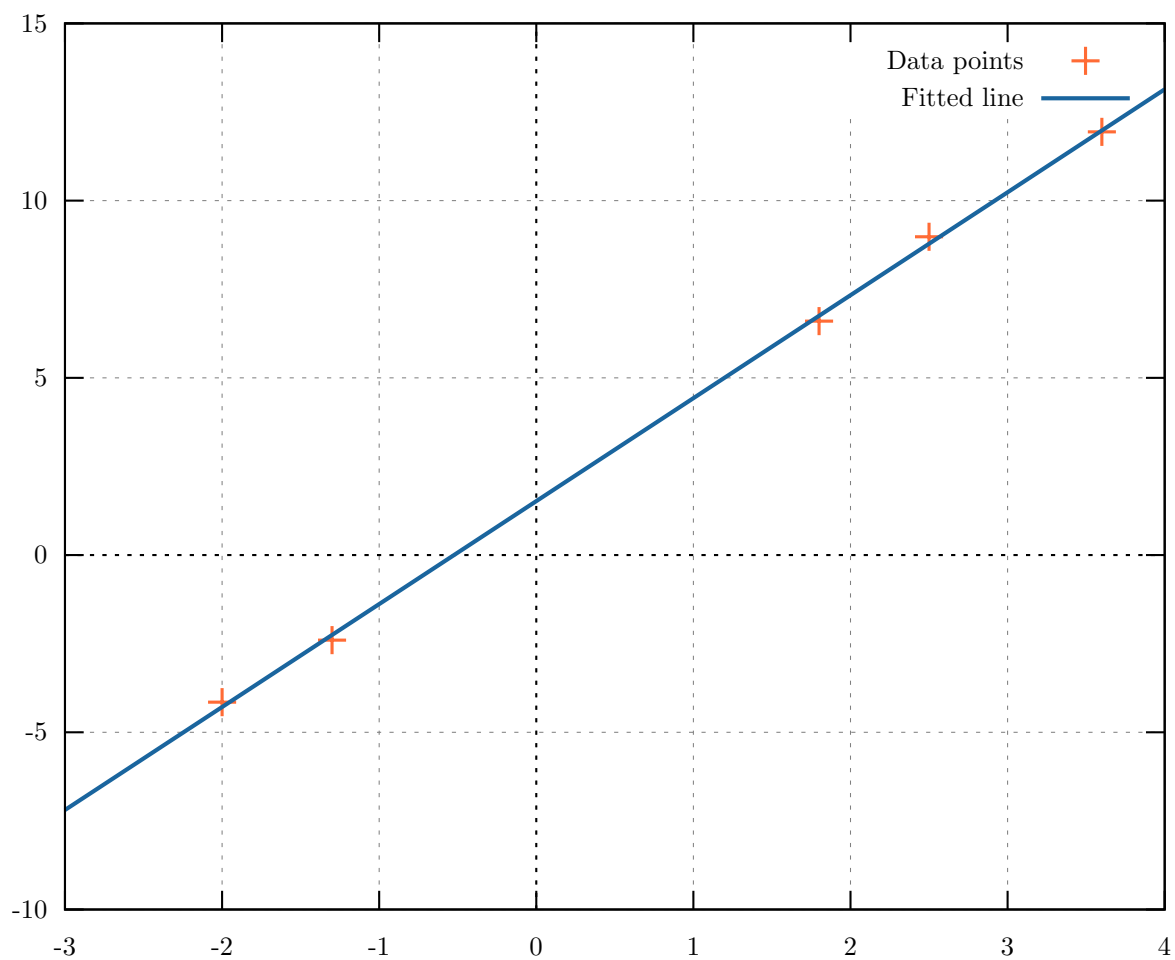


Figure 11: Linear fit for Problem Set 2 (iv).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 2.90453	+/- 0.03786	(1.304%)
c	= 1.52183	+/- 0.08983	(5.902%)

Question 5

Problem Set 2 (vi). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 3$. Plot the points and fitted line, show both axes, and choose the y -range so the intercept c is visible on the y -axis. Report m and c .

Table 5: Data for Problem Set 2 (vi).

x	y
-2.9	-5.96
-1.4	-3.81
0.8	-1.10
1.7	0.21
2.5	1.30

Data file contents:

```
-2.9 -5.96
-1.4 -3.81
0.8 -1.10
1.7 0.21
2.5 1.30
```

Sol: The data and fitted line are plotted below; the fit results follow the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_vi_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vi.log"
7
8 set grid
9 set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set xrange [-3:3]
13
14 y(x)=m*x+c
15
16 c=-2
17 m=1
18
19 fit y(x) "Lab-files/Data/2/Q_vi.txt" using 1:2 via m,c
20
21 plot "Lab-files/Data/2/Q_vi.txt" using 1:2 with p ps 1.5 lc "#f25c54" title "Data
    points", \
22     y(x) lw 2 lc "#f79d65" title "Fitted line"
```

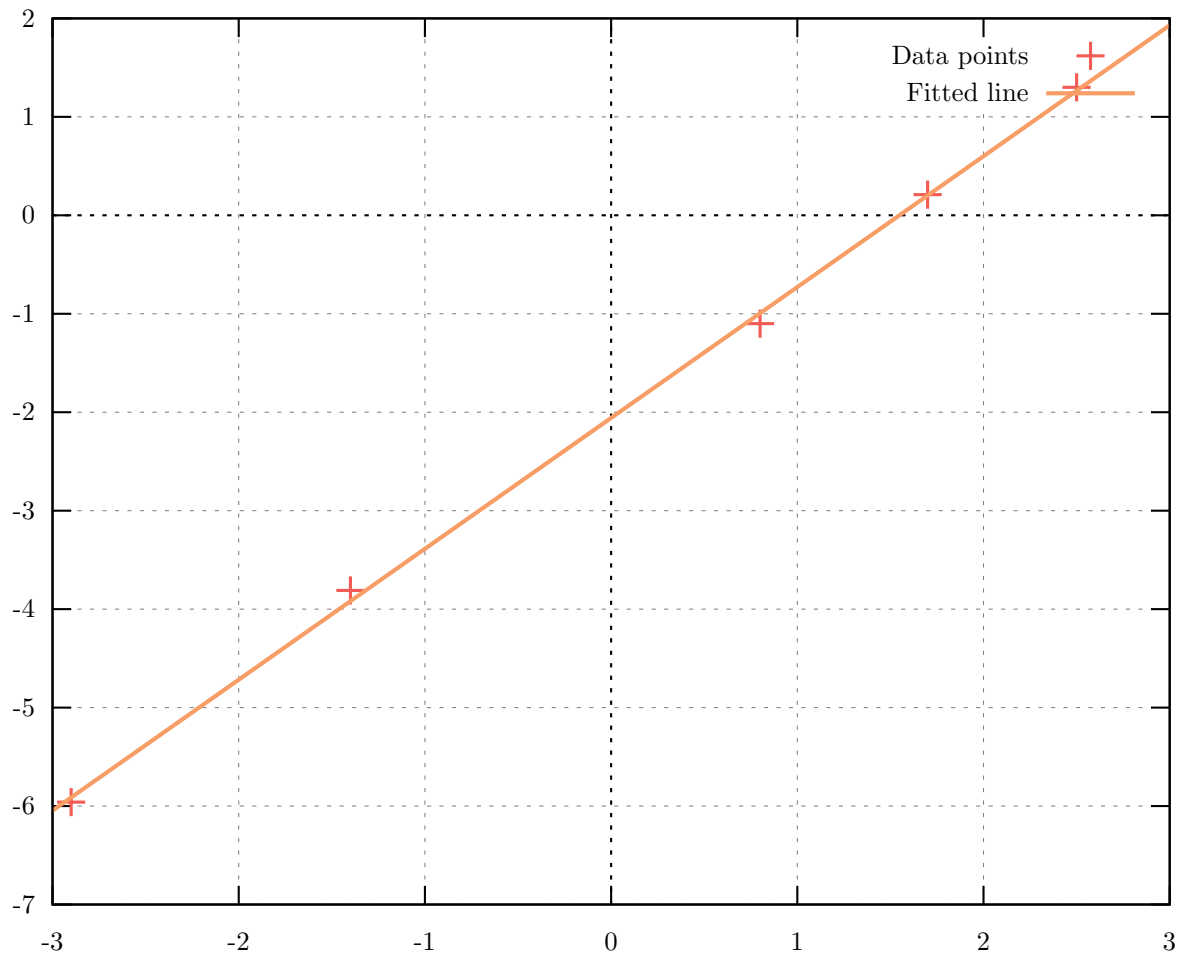


Figure 12: Linear fit for Problem Set 2 (vi).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 1.32931	+/- 0.02098	(1.579%)
c	= -2.0581	+/- 0.04213	(2.047%)

Question 6

Problem Set 2 (vii). Fit the data with $f(x) = ax^2 + b$ over $-3 \leq x \leq 3$ and $0 \leq y \leq 16$. Plot the points and fitted curve together, use the title *Polynomial fit*, and report a and b .

Table 6: Data for Problem Set 2 (vii).

x	y
-2.8	14.65
-1.2	5.81
0.7	4.48
1.8	8.33
2.9	15.56

Data file contents:

```
-2.8 14.65
-1.2 5.81
0.7 4.48
1.8 8.33
2.9 15.56
```

Sol: The data and fitted curve are plotted below; the fit results follow the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_vii_ou.svg"
4
5 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vii.log"
6 set title "Polynomial fit"
7 set grid
8 set xzeroaxis linestyle 0 lc "black"
9 set yzeroaxis linestyle 0 lc "black"
10
11 set xrange [-3:3]
12 set yrange [0:16]
13
14 y(x)=a*x**2+b
15
16 a=1
17 b=4
18
19 fit y(x) "Lab-files/Data/2/Q_vii.txt" using 1:2 via a,b
20 plot "Lab-files/Data/2/Q_vii.txt" using 1:2 with p ps 1.5 lc "#2b2d42" title "Data
    points", \
21     y(x) lw 2 lc "#d90429" title "Fitted curve"
```

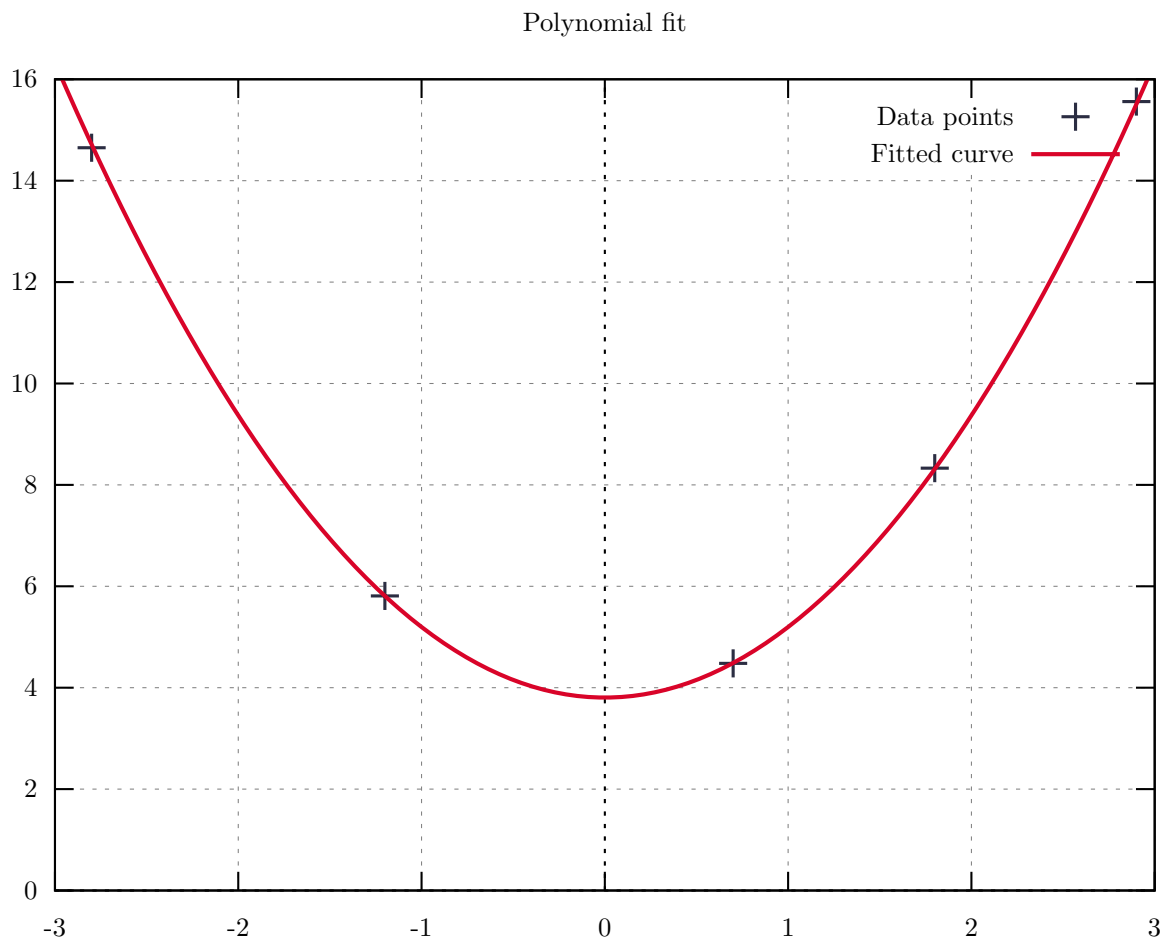


Figure 13: Polynomial fit for Problem Set 2 (vii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 1.39137	+/- 0.006722	(0.4831%)
b	= 3.80539	+/- 0.0362	(0.9513%)

Question 7

Problem Set 2 (ix). Fit the data with $f(x) = x^a$ and report a . Plot the data with the fitted function, title the plot *Data fitting*, and label the axes x and $f(x)$.

Table 7: Data for Problem Set 2 (ix).

x	$f(x)$
0.0	0.0
0.2	0.006
0.4	0.053
0.6	0.195
0.8	0.49
1.0	1.0
1.2	1.792
1.4	2.935
1.6	4.5
1.8	6.56
2.0	9.19

Data file contents:

```
0.0 0.0
0.2 0.006
0.4 0.053
0.6 0.195
0.8 0.49
1.0 1.0
1.2 1.792
1.4 2.935
1.6 4.5
1.8 6.56
2.0 9.19
```

Sol: The fitted power curve is plotted with the data points below; the fit result follows the plot.

```
1 reset
2 set terminal svg background "white"
3 set ou "Lab-files/Output/2/Q_2_ix_ou.svg"
4
5
6 set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_ix.log"
7
8 set grid
9 set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set title "Data fitting"
13 set xlabel "x"
14 set ylabel "f(x)"
15
16 y(x)=x**a
17
18 fit y(x) "Lab-files/Data/2/Q_ix.txt" using 1:2 via a
19 plot "Lab-files/Data/2/Q_ix.txt" using 1:2 with p ps 1.5 lc "#a00000" title "Data
    points", \
20     y(x) lw 2 lc "#f66420" title "Fitted curve"
```

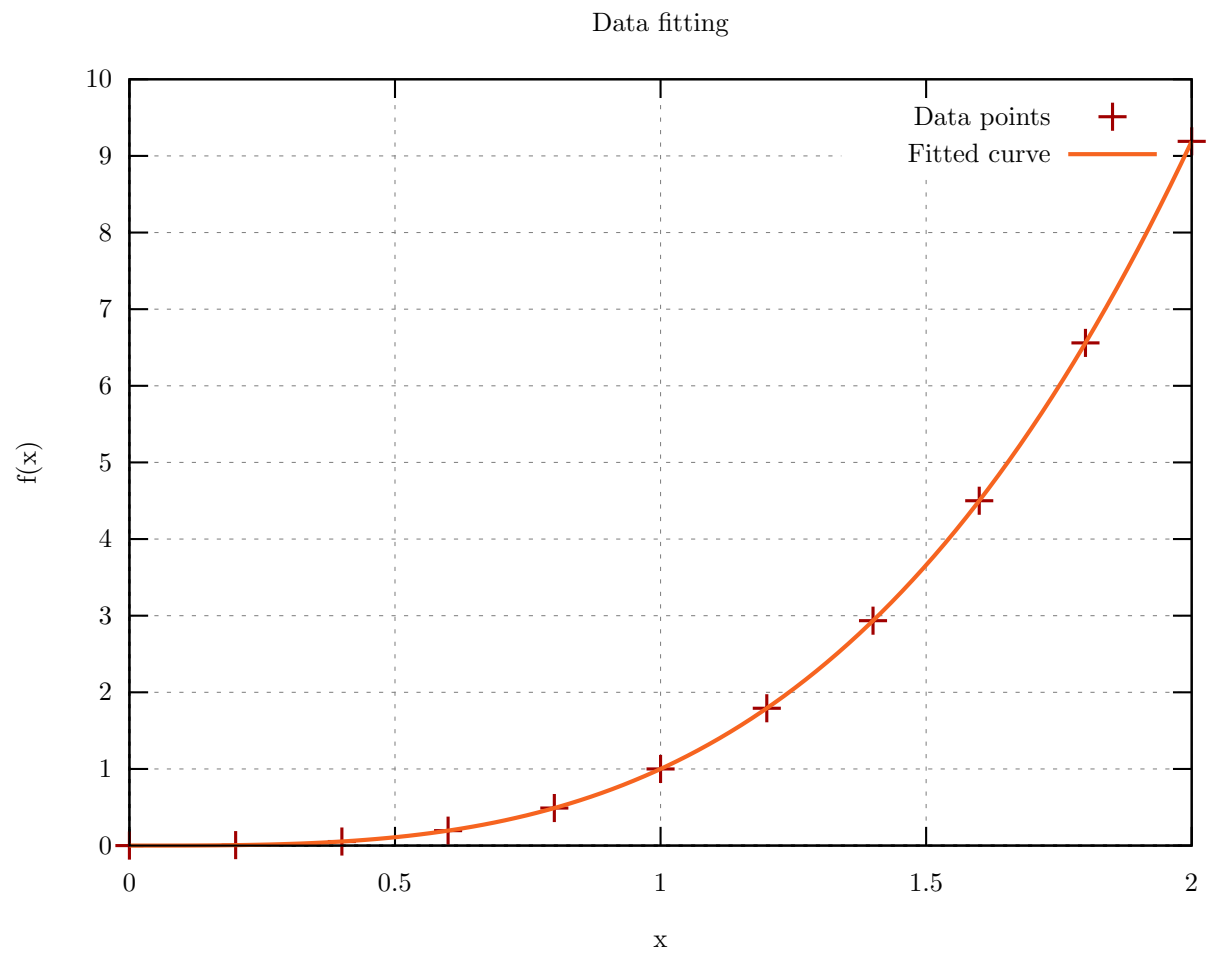


Figure 14: Power fit for Problem Set 2 (ix).

Fit log (parameter estimate and standard error):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 3.20008	+/- 2.388e-05	(0.0007463%)

Section 3

Question 1

Problem Set 3 (i). Show the polar plots of $\sin^3(t)$ and $\cos^2(2t)$. Label the axes “t value” and “the range”, place “polar plot” at $(0, -0.7)$, and show the vertical axis. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set polar
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3_i_ou.svg"
6
7 set title "Polar plot"
8 set grid polar linestyle 0 lc "black"
9 set xlabel "t value"
10 set ylabel "the range"
11 set label "polar plot" at 0,-0.7
12 set yzeroaxis linestyle 1 lc "black"
13
14 set samples 1000
15 plot (sin(t))**3 lw 2 lc "#772f1a",cos(2*t)**2 lw 2 lc "#f2a65a"
```

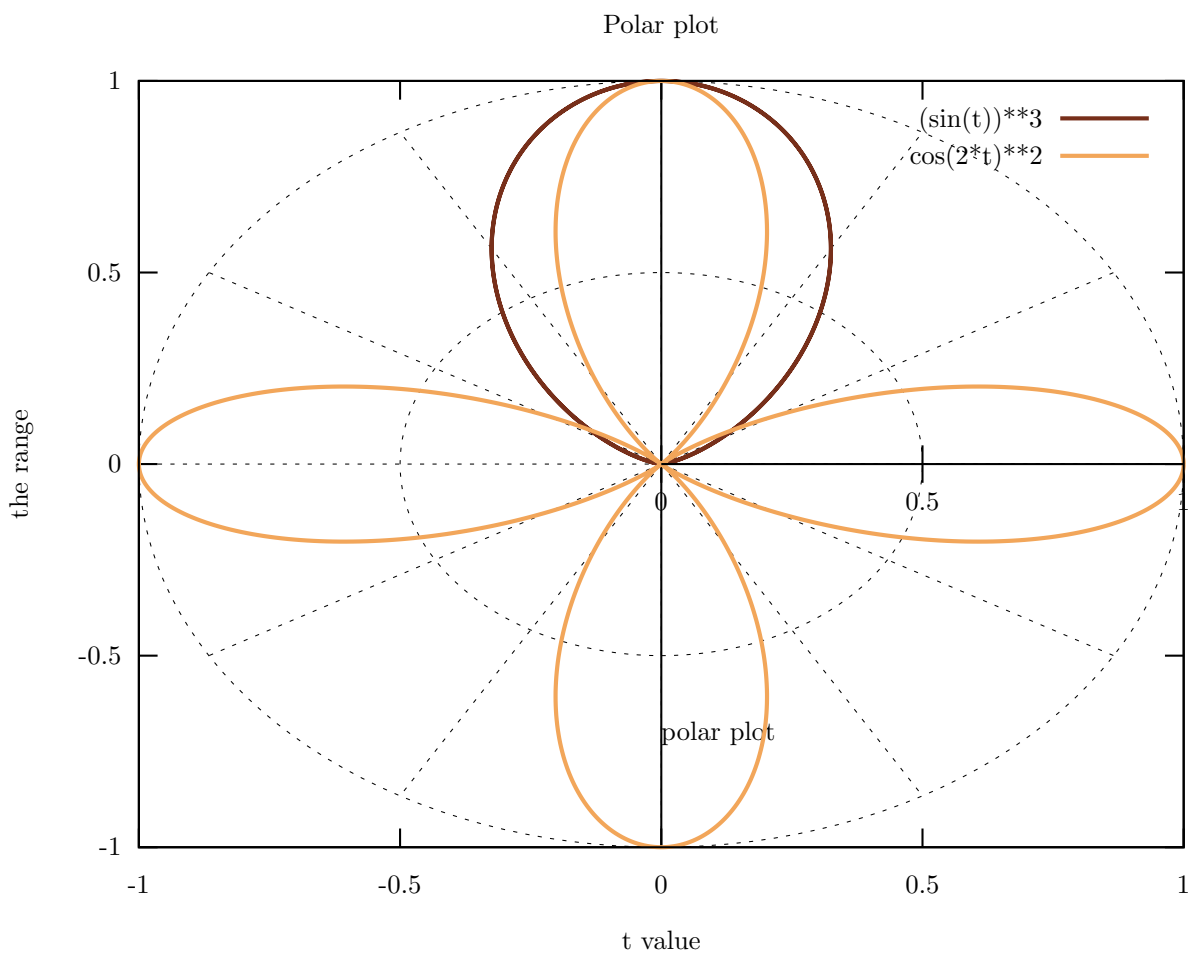


Figure 15: Polar plots of $\sin^3(t)$ and $\cos^2(2t)$.

Question 2

Problem Set 3 (ii). Show the polar plots of $\tan(6t)$ and $\cot(6t)$ over ranges -20 to 20 on both axes. Place the label “ $\tan(6t)$, $\cot(6t)$ ” at $(4, 12)$ and show both axes. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set polar
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3_ii_ou.svg"
6
7 set title "Polar plot"
8
9 set grid polar linestyle 1
10 set xlabel "t value"
11 set ylabel "the range"
12 set xzeroaxis linestyle 1 lc "black"
13 set yzeroaxis linestyle 1 lc "black"
14 set label "tan(6t), cot(6t)" at 4,12
15
16 set xrange [-20:20]
17 set yrange [-20:20]
18
19
20 set samples 1000
21
22 plot tan(6*t) lc "#780116", (1/tan(6*t)) lc "#e85d04"
```

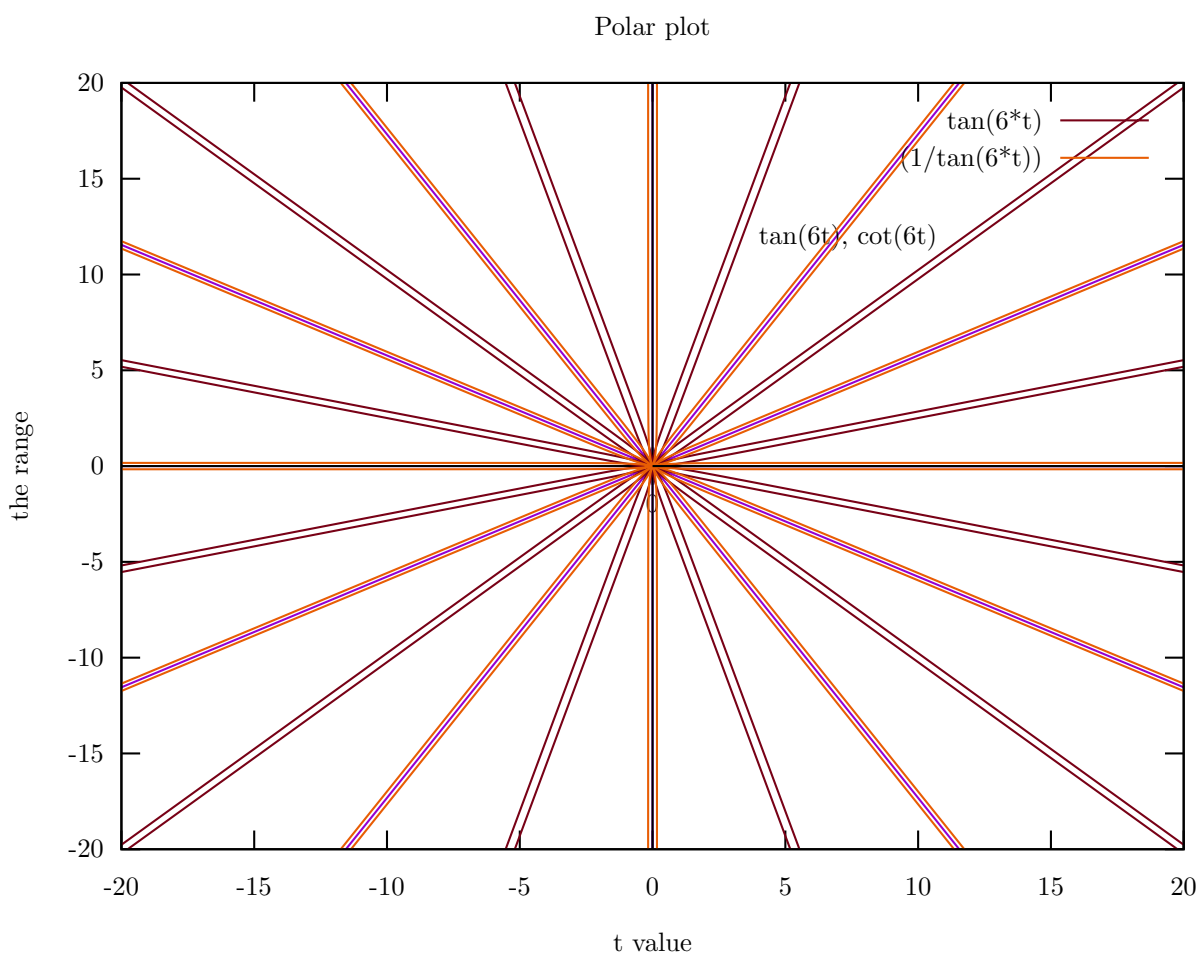


Figure 16: Polar plots of $\tan(6t)$ and $\cot(6t)$.

Question 3

Problem Set 3 (iii). Plot the parametric curve $x = 2 \cos^3 t$, $y = 2 \sin^3 t$ for $-\pi \leq t \leq \pi$. Label the axes as specified and title the plot “Astroid”. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set parametric
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3_iii_ou.svg"
6
7 set title "Astroid"
8 set xlabel "x = 2 cos(t)^3"
9 set ylabel "y = 2 sin(t)^3"
10 set grid linestyle 1 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16 x(t)=2*cos(t)**3
17 y(t)=2*sin(t)**3
18
19 set samples 1000
20
21 plot x(t),y(t) lw 2 lc "#dc2f02"
```

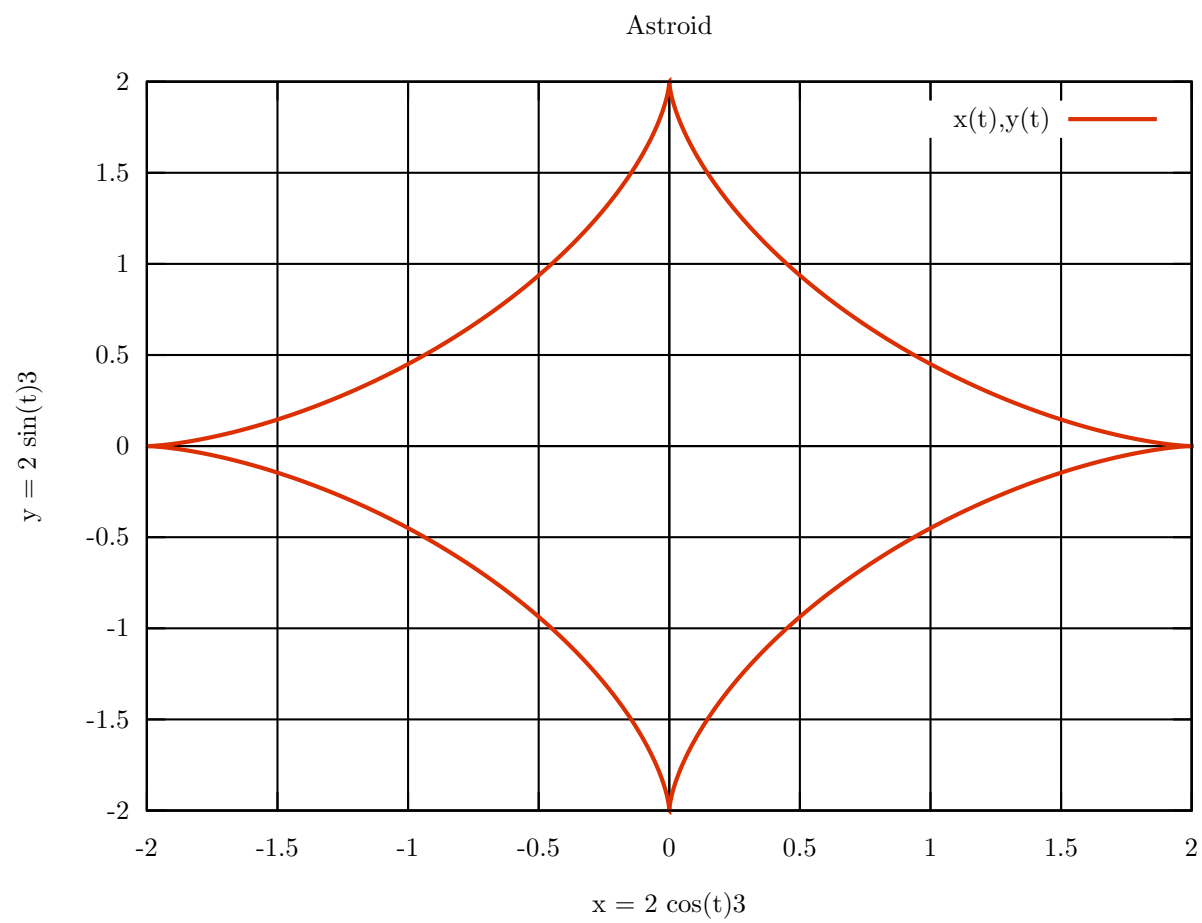


Figure 17: Astroid parametric plot.

Question 4

Problem Set 3 (iv). Plot the polar spiral $r = t$ for $0 \leq t \leq 6\pi$ using 10000 samples. Show both axes, label them as specified, and title the plot “Spiral”. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set polar
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3_iv_ou.svg"
6
7 set title "Spiral"
8 set xlabel "x = r cos t"
9 set ylabel "y = r sin t"
10 set grid polar lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [0:6*pi]
16 r(t)=t
17
18 set samples 10000
19
20 plot r(t) lw 2 lc "#9d0208"
```

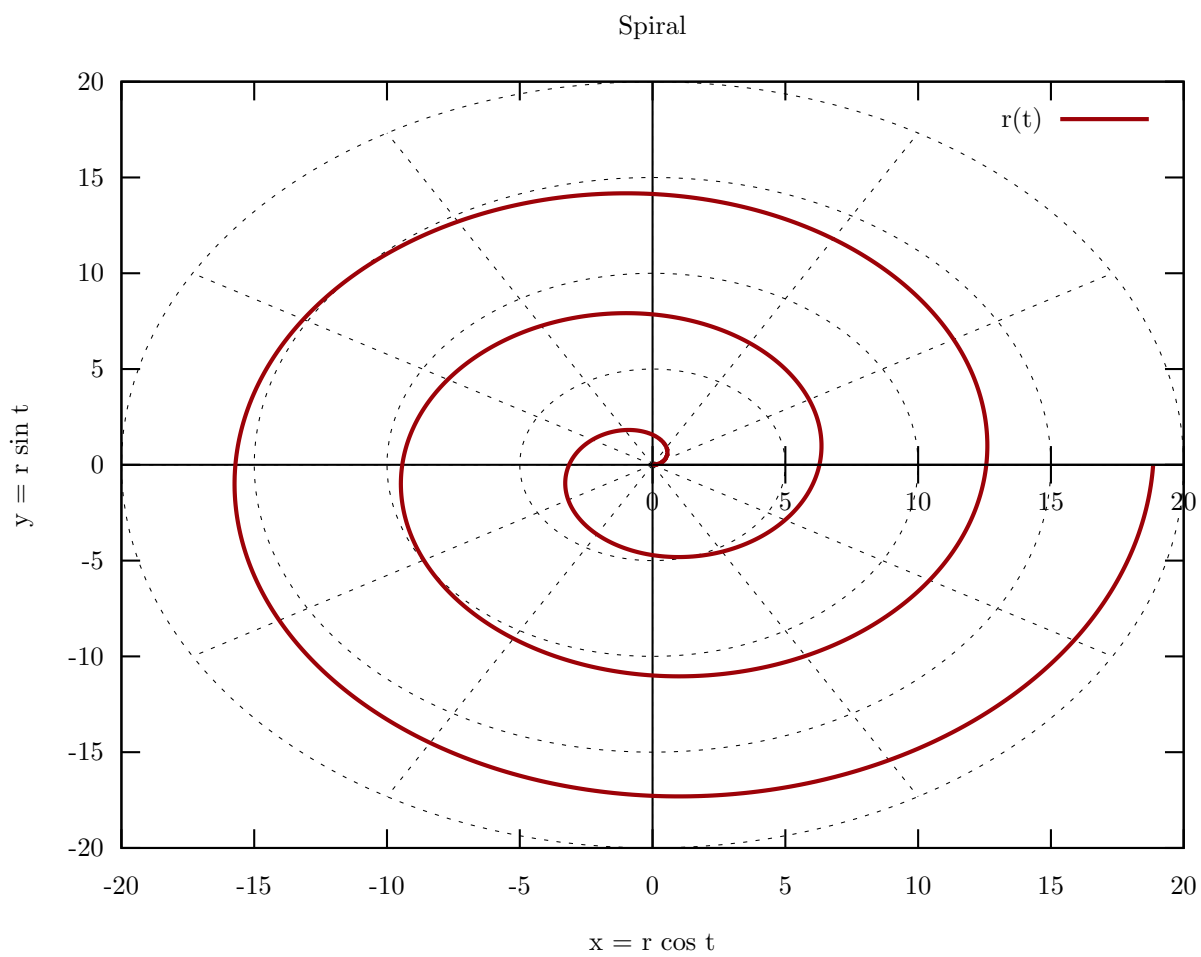


Figure 18: Polar spiral $r = t$.

Question 5

Problem Set 3 (v). Plot the parametric curve $x = 2 \sin t$, $y = 2 \sin(2t)$ for $-\pi \leq t \leq \pi$, with the specified axis labels and title “Lissajous Figure”. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set parametric
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3_v_ou.svg"
6
7 set title "Lissajous Figure"
8 set xlabel "x = 2 sin t"
9 set ylabel "y = 2 sin 2t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16
17 x(t)=2*sin(t)
18 y(t)=2*sin(2*t)
19
20
21 set samples 1000
22
23 plot x(t),y(t) lw 2 lc "#e36414"
```

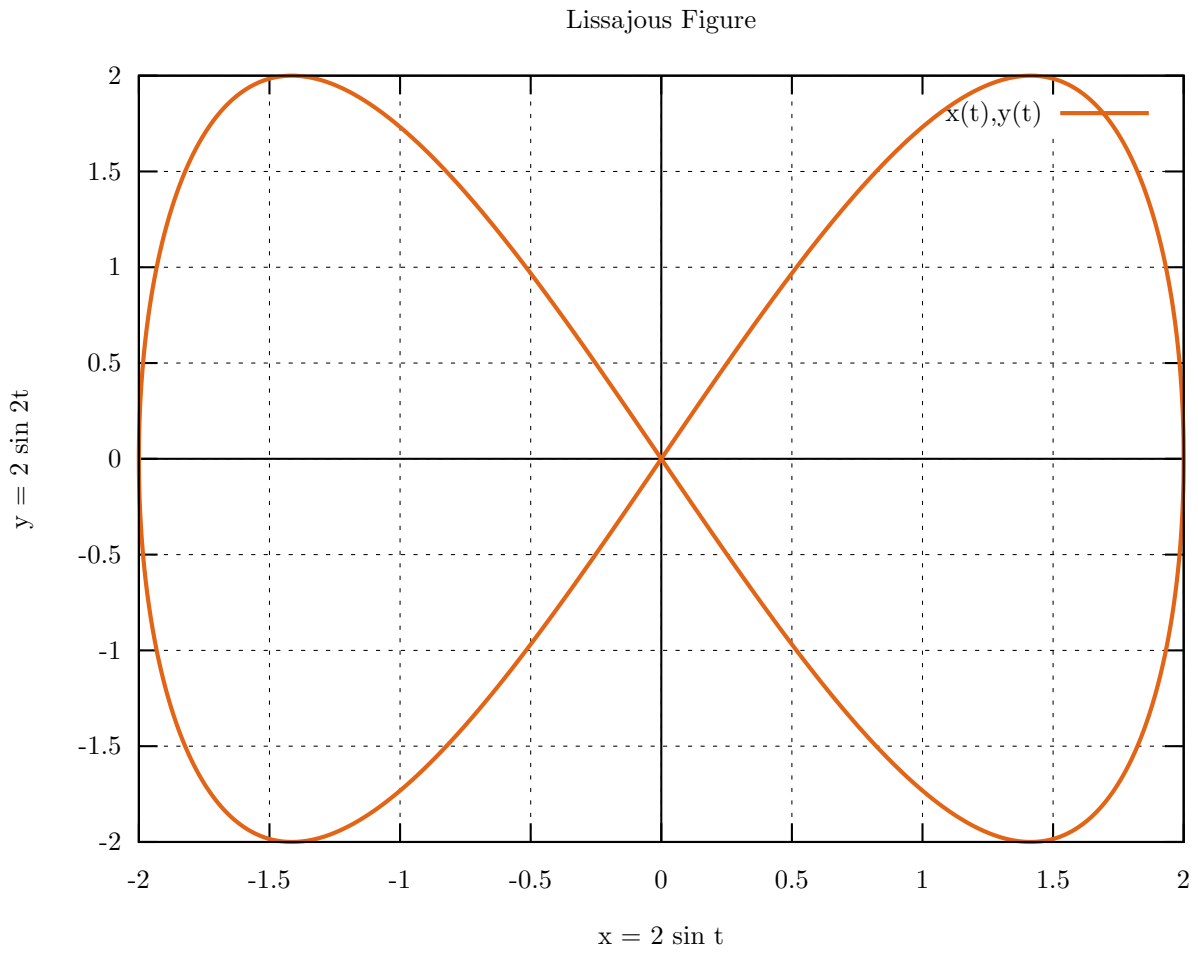


Figure 19: Lissajous parametric plot.

Question 6

Problem Set 3 (vi). Plot the polar curves $r_1 = 4 + 3 \cos t$ and $r_2 = 3 + 4 \cos t$. Label the axes as specified, place “polar plot” at $(-6, 6)$, and title the plot “A pair of Limacon”. **Sol:** The corresponding Gnuplot program and output are below.

```
1 reset
2
3 set polar
4 set terminal svg background "white"
5 set ou "Lab-files/Output/3/Q_3-vi_ou.svg"
6
7 set title "A pair of Limacon"
8 set xlabel "x = r cos t"
9 set ylabel "y = r sin t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13 set label "polar plot" at -6,6
14
15 set trange [-pi:pi]
16
17 r1(t)=4+3*cos(t)
18 r2(t)=3+4*cos(t)
19
20
21 set samples 1000
22
23 plot r1(t) lw 2 lc "#5f0f40", r2(t) lw 2 lc "#e36414"
```

A pair of Limacon

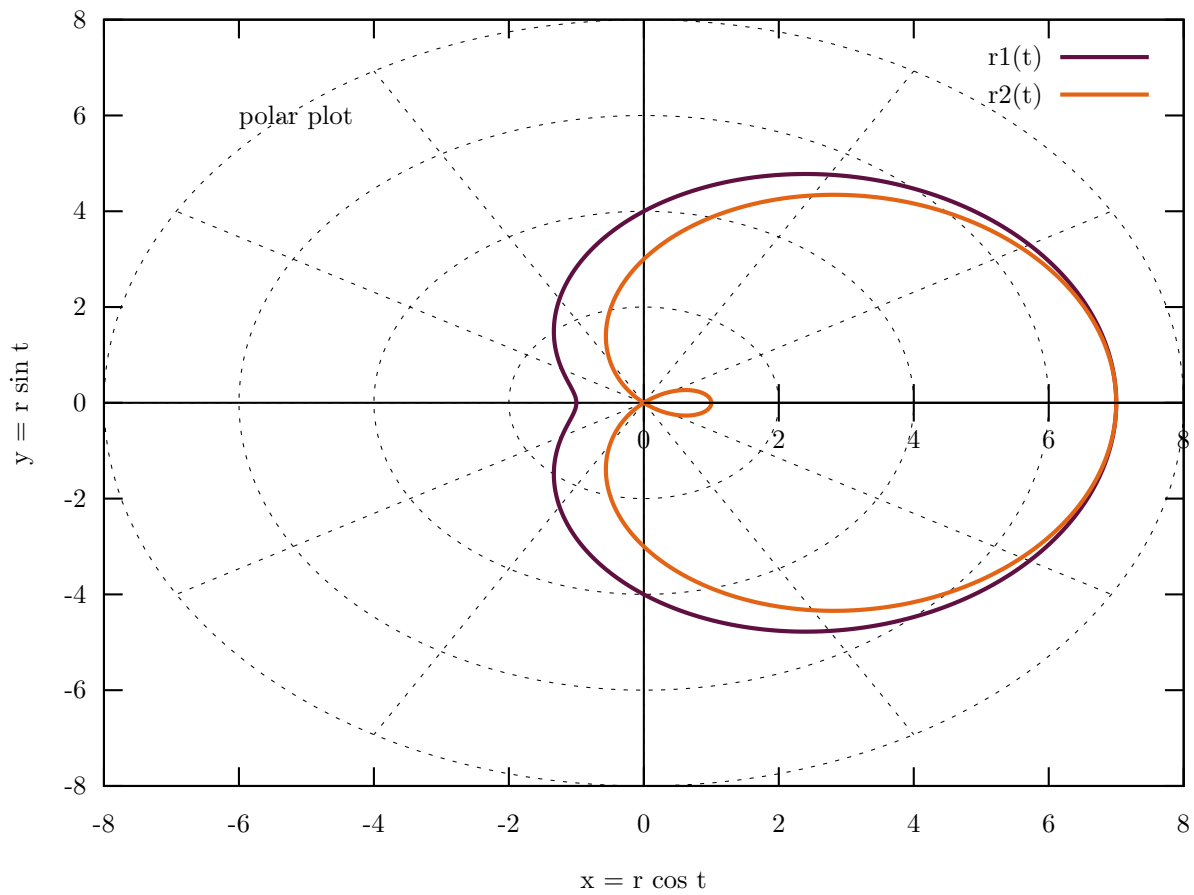


Figure 20: A pair of limaçons.